

Lecture Notes

EC226 Term 1 2025/26

Two variable linear regression analysis

Readings:

- Stock and Watson (2003) Chapters 4 + 5
- Dougherty (2016) Chapter 2
- Wooldridge (2013) Chapter 2

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1 Introduction

Econometrics means economic measurement. Hendry (1980) described econometrics as “An analysis of the relationship between economic variables ... by abstracting the main phenomena of interest and stating theories thereof in mathematical form” , and Samuelson et al. (1954) stated that “Econometrics may be defined as the quantitative analysis of actual economic phenomena” Other authors have been less complementary, Leamer (1983) believes that econometrics “is practised at the computer terminal (and) involves fitting many, perhaps thousands, of statistical models. One or several that the researcher finds pleasing are selected for reporting purposes”.

Econometrics has three major uses:

- 1 Describing economic reality.
- 2 Testing hypotheses about economic theory.
- 3 Forecasting future economic activity.

Econometricians attempt to quantify economic relationships that had previous been only theoretical. To undertake this requires 3 steps:

1. Specifying/identifying theoretical economic relationship between the variables.
2. Collecting the data on those variables identified by the theoretical model.
3. Obtaining estimates of the parameters in the theoretical relationship.

2 Correlation vs Regression

2.1 Correlation

In Economics we are interested in the relation between 2 or more random variables, for example:

- Sales and advertising expenditure
- Personal consumption and disposable income
- Investment and interest rates
- Earnings and schooling

While there are many ways in which these pairs of variables might be related – a linear relationship is often a useful first approximation and this can be detected via a scatter plot, of one variable against the other.

A measure of linear association between two random variables x and y is the covariance, which for a sample of n pairs of observations $(x_1, y_1) \dots (x_n, y_n)$ is calculated as:

$$cov(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

The covariance measures the average cross product of deviations of x , around its mean, with y , around its mean. If high (low) values of x (relative to its mean) are associated with high (low) values of y (relative to its mean) then we get a high positive covariance (Figure 1). Conversely if high (low) values of x (relative to its mean) are associated with low (high) values of y (relative to its mean) we get a negative covariance (Figure 2). A zero covariance occurs when there is no predominant association between the x and y values (Figure 3). The covariance is a linear association between x and y values and would be approximately zero for a quadratic association (Figure 4).

Figure 1: Positive correlation between y and x

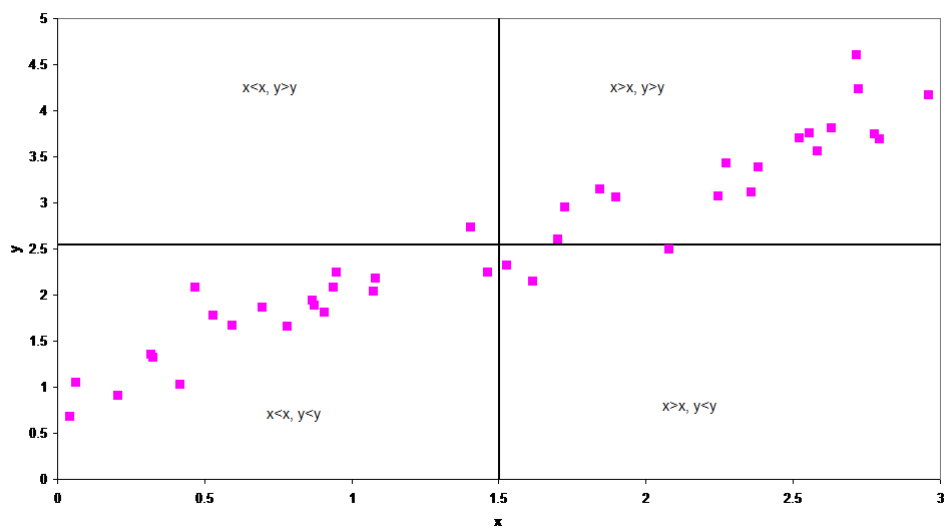


Figure 2: Negative correlation between y and x

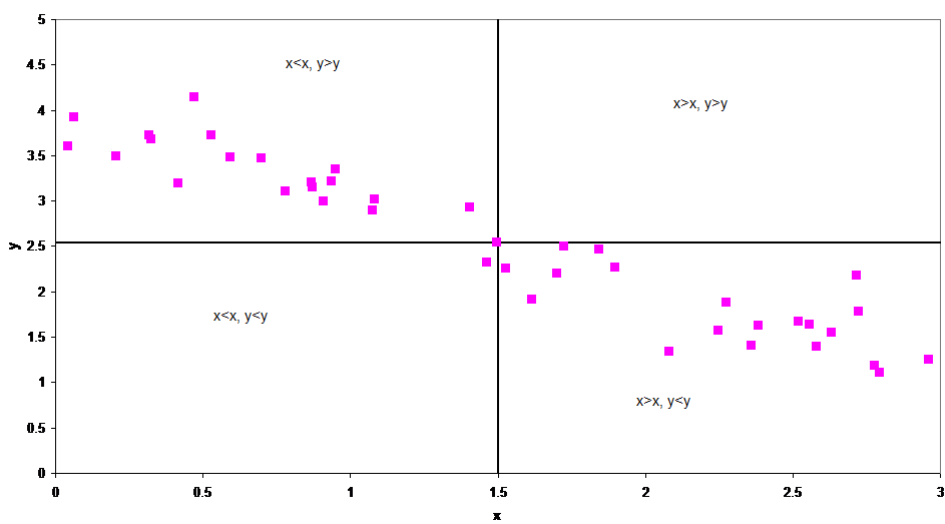


Figure 3: Zero correlation between y and x

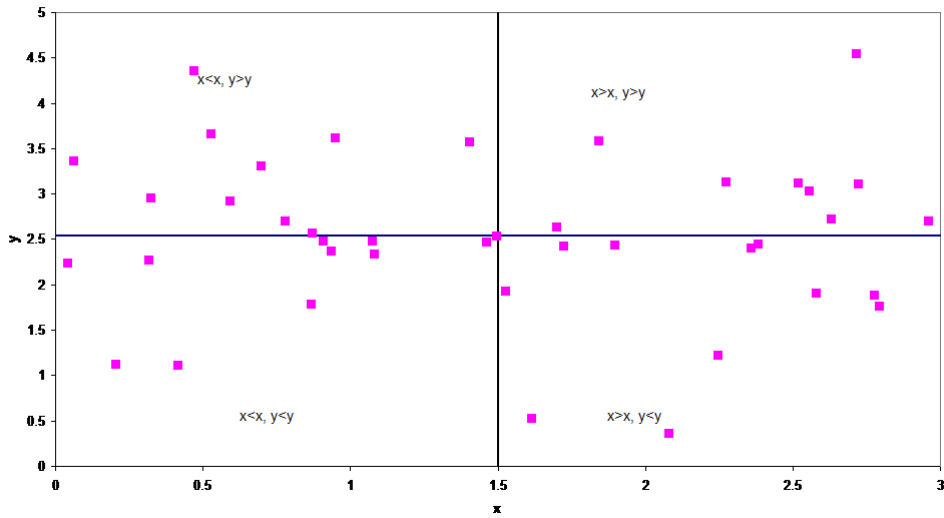
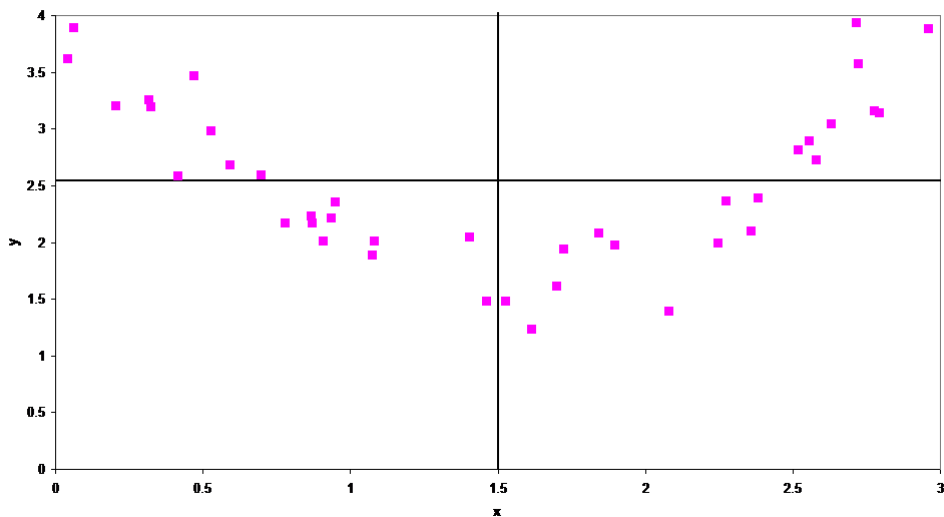


Figure 4: Zero correlation between y and x



The covariance is not scale free statistic and multiplying the x variable by 100 multiplies the covariance by 100. A scale free measure is a correlation, which is defined as:

$$\text{corr}(x, y) \equiv \rho(x, y) = \frac{\text{cov}(x, y)}{\sqrt{V(x)V(y)}}$$

this is scale free because

$$\text{corr}(ax, y) = \frac{\text{cov}(ax, y)}{\sqrt{V(ax)V(y)}} = \frac{a \cdot \text{cov}(x, y)}{\sqrt{a^2 V(x)V(y)}} = \frac{a \cdot \text{cov}(x, y)}{a \sqrt{V(x)V(y)}} = \text{corr}(x, y)$$

$\rho(x, y)$ is a population parameter of association between the random variables x and

y , and:

1. $-1 \leq \rho(x, y) \leq 1$
2. $\rho(x, y) = -1 \Rightarrow$ perfect negative association
3. $\rho(x, y) = 1 \Rightarrow$ perfect positive linear association
4. $\rho(x, y) = 0 \Rightarrow$ no linear association
5. As $|\rho(x, y)|$ increases $\Rightarrow$ stronger association.
6. $\rho(x, y) = \rho(y, x)$

2.2 Regression

By contrast linear regression looks at the linear causal association between the random variables x and y . In particular, we talk about the variable, x , taking a specific value and we are interested in the response of y to a change in this value of x . So in our examples above we might be interested in

- Changes in sales caused by increased advertising expenditure
- Changes in personal consumption caused by increased disposable income
- Changes in investment caused by increased interest rates
- Changes in earnings caused by increased schooling

In the simplest type of linear regression analysis we model the relationship between 2 variables, y and x , and this is assumed to be a linear relationship. In particular, we are interested in the expected value of the random variable, y , given a specific value for x . Given linearity this is ¹:

$$E(y|x) = \alpha + \beta x$$

where

$\alpha = E(y|x = 0) \Rightarrow$ expected value of y when $x=0$ (invariably do not interpret this)

$$E(y|x + 1) = \alpha + \beta(x + 1)$$

$\beta = E(y|x + 1) - E(y|x) \Rightarrow$ change in the expected value of y for a unit increase in x .
 y – is known as the **dependent variable** (**endogenous variable** or **regressand**)

¹Appendix A has some rules on expectations and variances.

x – is known as the **independent variable** (**exogenous variable**, **explanatory variable** or **regressor**).

The actual values of the dependent variable, y , will not be the same as the expected value and we denote the discrepancy between the actual and expected value by ε_i , where:

$$\varepsilon_i = y_i - E(y_i|x_i) = y_i - (\alpha + \beta x_i)$$

Rearranging we have

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad i = 1, 2, \dots, n \quad (1)$$

and this is the TRUE (but unknown) relationship between y and x and is made up of two components:

- 1 $\alpha + \beta x_i$ - the systematic part
- 2 ε_i - the random (non-systematic) component (error or disturbance).

3 CLRM assumptions

To complete the model we need to specify the statistical properties of x and ε and these are known as the Classical Linear Regression Model (CLRM) assumptions:

- 1 $E(\varepsilon_i|x_i) = E(\varepsilon_i) = 0 \forall i$ (so the error term is independent of x_i).
- 2 $V(\varepsilon_i|x_i) = \sigma^2 \forall i$ (error variance is constant (homoscedastic) – points are distributed around the true regression line with a constant spread).
- 3 $cov(\varepsilon_i, \varepsilon_j|x_i) = 0$ for $i \neq j$ (the errors are serially uncorrelated over observations).
- 4 $\varepsilon_i|x_i \sim N(0, \sigma^2) \forall i$.

Figure 5 plots the TRUE (unknown) regression line and the distribution of points around this which are consistent with the CLRM assumptions. Figure 6 plots the actual TRUE (unknown) disturbance terms from Figure 5, which are consistent with the CLRM assumptions.

Figure 5: The two variable regression model

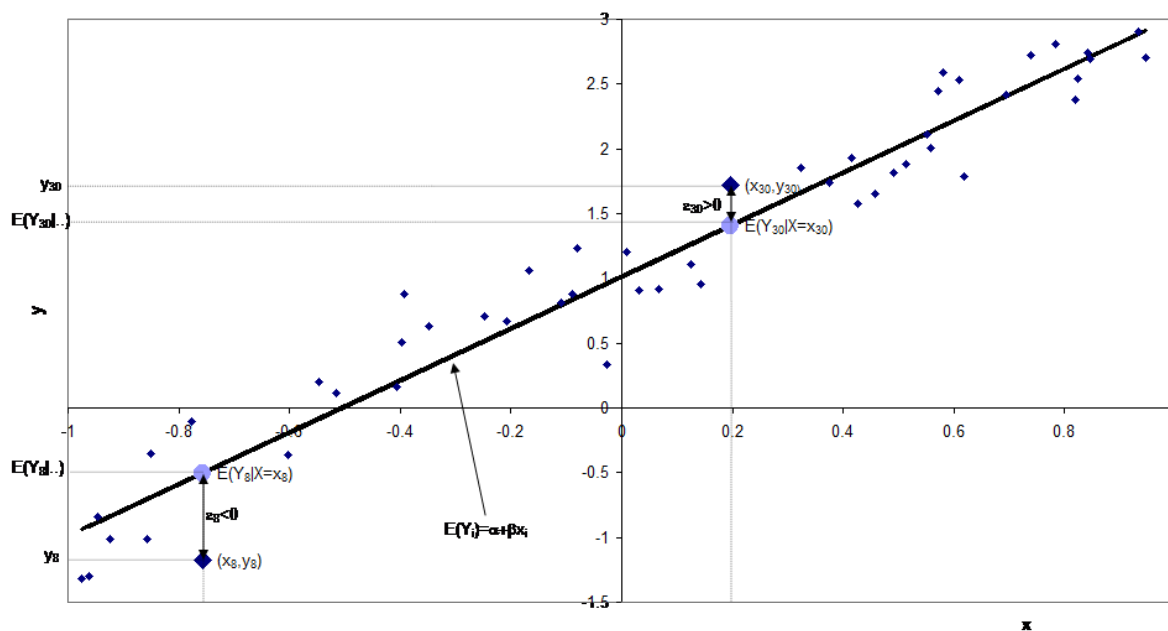
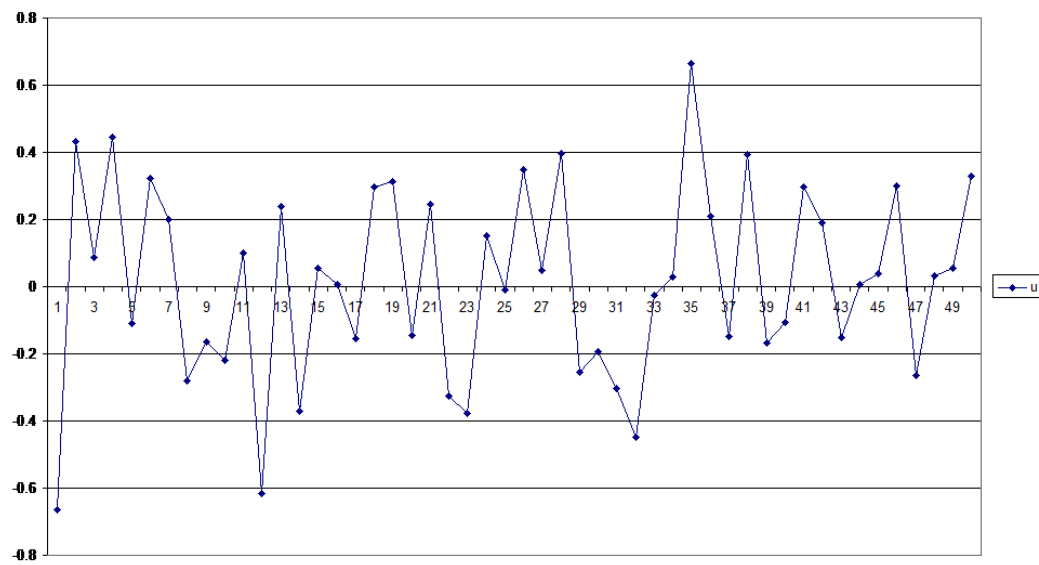


Figure 6: Disturbances which satisfy the CLRM assumptions

4 Important questions

- 1 How do we estimate the population parameters $(\alpha, \beta, \sigma^2)$ of this statistical model?
- 2 What are the properties of the estimators?
- 3 How do we test hypotheses about these parameters?
- 4 How strong is the relationship?
- 5 Is the model adequate?
- 6 Can the model be used for forecasting?

4.1 Estimation of the sample regression line

The statistical model is:

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad i = 1, 2, \dots, n$$

- 1 $E(\varepsilon_i | x_i) = 0$.
- 2 $V(\varepsilon_i | x_i) = \sigma^2$.
- 3 $cov(\varepsilon_i, \varepsilon_j | x_i) = 0$.
- 4 $\varepsilon_i | x_i \sim N(0, \sigma^2)$.

We are interested in estimating the population parameters, α, β, σ^2 , we will call the estimates, a, b and s^2 .

Note: For ease of notation we are not going to explicitly write conditional on x_i .

For n pairs of observations $(x_1, y_1) \dots (x_n, y_n)$ we want to find the straight line that best fits these points, denote:

$$\hat{y}_i = a + bx_i$$

as the predicted values of y from the model. In which case we can define

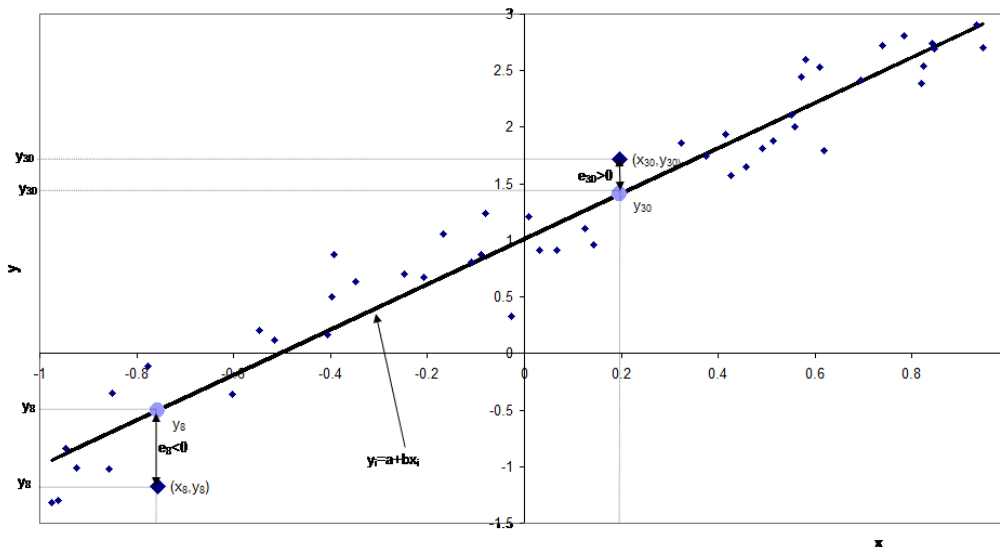
$$y_i = \hat{y}_i + e_i = a + bx_i + e_i$$

and e are the residuals (difference between the actual value of y and its predicted value, $\hat{y}_i$), such that,

$$e_i = y_i - \hat{y}_i$$

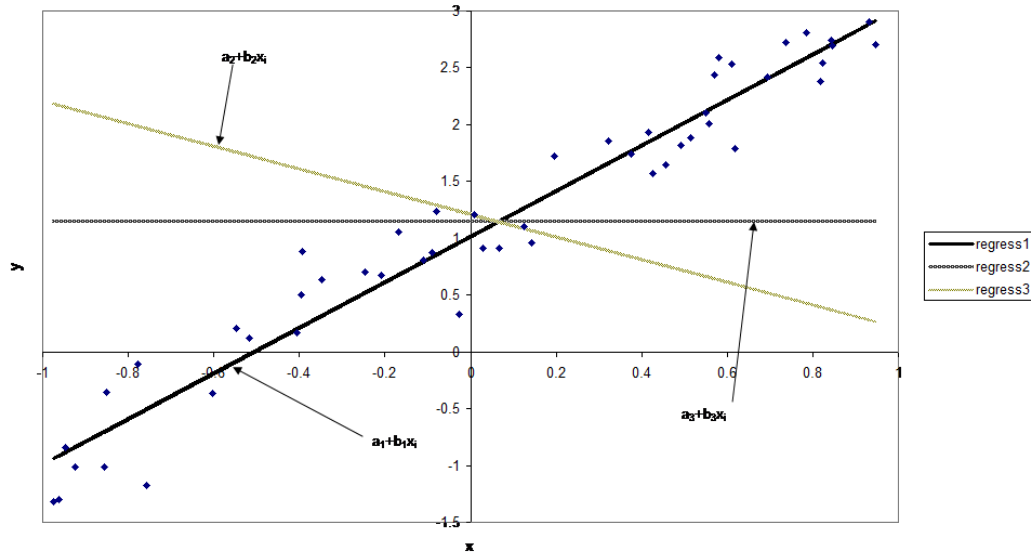
see Figure 7 for an illustration of the data points (black diamonds), the predicted line (black line) and the residuals (vertical distance between the diamonds and the predicted line) and highlighted for e_8 and e_{30} .

Figure 7: The estimated two variable regression model



For any given values of a and b we can define a regression line (in Figure 8 we plot three alternative regression lines for a_i and b_i $i = 1, 2, 3$). But we want a and b to have some desirable properties. The best line is that which makes the residuals, e_i , as small as possible. However, as residuals can be both positive and negative, obtaining lines such that $\sum_{i=1}^n e_i = 0$ can yield a variety of equally good lines (in Figure 8 all three lines have this property).

Figure 8: Two-variable regression - line of best fit



The optimal solution is to minimise the RSS (Residual Sum of Squares)

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

with respect to the two unknown parameters a and b .

We therefore differentiate the RSS with respect to a and b and set the resultant expressions equal to zero – this process of minimising the RSS to obtain the parameter estimates is known as Ordinary Least Squares (OLS).

Differentiating the RSS with respect to the parameter, a , and setting the expression to zero:

$$\frac{\partial \sum_{i=1}^n (y_i - a - bx_i)^2}{\partial a} = \frac{\partial [(y_1 - a - bx_1)^2 + \dots + (y_n - a - bx_n)^2]}{\partial a} = 0 \quad (2)$$

This entails differentiating each of the n terms in equation (2) with respect to a :

$$\begin{aligned} \frac{\partial (\sum_{i=1}^n (y_i - a - bx_i)^2)}{\partial a} &= -2(y_1 - a - bx_1) - \dots - 2(y_n - a - bx_n) = 0 \\ &= -2 \sum_{i=1}^n \underbrace{(y_i - a - bx_i)}_{e_i} = -2 \sum_{i=1}^n e_i = 0 \Rightarrow \sum_{i=1}^n e_i = 0 \end{aligned} \quad (3)$$

Differentiating the RSS with respect to the parameter, b , and setting the expression

to zero:

$$\frac{\partial(\sum_{i=1}^n (y_i - a - bx_i)^2)}{\partial b} = \frac{\partial [(y_1 - a - bx_1)^2 + \dots + (y_n - a - bx_n)^2]}{\partial b} = 0 \quad (4)$$

differentiating each of these n terms in equation (4) with respect to b :

$$\begin{aligned} \frac{\partial(\sum_{i=1}^n (y_i - a - bx_i)^2)}{\partial b} &= -2x_1(y_1 - a - bx_1) - \dots - 2x_n(y_n - a - bx_n) = 0 \\ &= -2 \sum_{i=1}^n x_i \underbrace{(y_i - a - bx_i)}_{e_i} = -2 \sum_{i=1}^n x_i e_i = 0 \Rightarrow \sum_{i=1}^n x_i e_i = 0 \end{aligned} \quad (5)$$

NOTE

- (i) Equation (3) implies that the residuals always sum to zero (providing there is an intercept in the model)
- (ii) Equation (5) implies that the covariance (and hence correlation) between the residuals and the x 's is zero, that is, they are orthogonal.

The two equations we obtained from (3) and (5) which are: $\sum_{i=1}^n e_i = 0$ and $\sum_{i=1}^n x_i e_i = 0$ are called **NORMAL equations**.

Appendix G has a small empirical example of estimating a two variable regression model by OLS, where we show that $\sum_{i=1}^n e_i = 0$ and $\sum_{i=1}^n x_i e_i = 0$.

Solving our NORMAL equation (3) for a we have:

$$\sum_{i=1}^n e_i \equiv \sum_{i=1}^n y_i - \sum_{i=1}^n a - \sum_{i=1}^n bx_i = 0 \Rightarrow \sum_{i=1}^n y_i - na - b \sum_{i=1}^n x_i = 0$$

which implies:

$$na = \sum_{i=1}^n y_i - b \sum_{i=1}^n x_i \Rightarrow a = \bar{y} - b\bar{x} \quad (6)$$

Substituting out a in equation (5) using equation (6):

$$\begin{aligned} \Rightarrow \sum_{i=1}^n x_i(y_i - (\bar{y} - b\bar{x}) - bx_i) &= 0 \Rightarrow \sum_{i=1}^n x_i(y_i - \bar{y}) - b \sum_{i=1}^n x_i(x_i - \bar{x}) = 0 \\ b \sum_{i=1}^n x_i(x_i - \bar{x}) &= \sum_{i=1}^n x_i(y_i - \bar{y}) \\ b &= \frac{\sum_{i=1}^n x_i(y_i - \bar{y})}{\sum_{i=1}^n x_i(x_i - \bar{x})} \equiv \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned} \quad (7)$$

We also need to estimate σ^2 (the variance of the disturbance term, ε_i), this is estimated as:

$$s^2 = \frac{\sum_{i=1}^n e_i^2}{DoF} = \frac{RSS}{DoF} \quad (8)$$

where DoF are the degrees of freedom for the residuals. The DoF is the number of observations, n , less the number of restrictions we have on these residuals and these restrictions come from the two NORMAL equations, that is, $\sum_{i=1}^n e_i = 0$ and $\sum_{i=1}^n x_i e_i = 0$ (which is equal to the number of estimated parameters, a and b), i.e. $n - 2$.

These OLS estimates have a number of desirable properties (known as the Gauss Markov Theorem) in that the estimators are **BLUE**:

- 1 Best - the OLS estimators have the minimum variance, such that $V(b|x) \leq V(b^*|x)$, where b^* is any alternative unbiased estimator.
- 2 Linear - the OLS estimators are a linear function of the error term.
- 3 Unbiased - the OLS estimators are such that: $E(a|x) = \alpha$ and $E(b|x) = \beta$
- 4 Estimators.

4.2 Properties of the OLS estimators

From equation (1) taking averages we have that $\bar{y} = \alpha + \beta\bar{x} + \bar{\varepsilon}$, in which case we can write equation (1) as:

$$y_i - \bar{y} = \beta(x_i - \bar{x}) + \varepsilon_i - \bar{\varepsilon} \quad (9)$$

substituting equation (9) into equation (7) we have:

$$b = \frac{\sum_{i=1}^n (x_i - \bar{x}) \{ \beta(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon}) \}}{\sum_{i=1}^n (x_i - \bar{x})^2} = \beta \frac{\overbrace{\sum_{i=1}^n (x_i - \bar{x})^2}^1}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (10)$$

Defining $\omega_i = \frac{(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ where

$$(i) \sum_{i=1}^n \omega_i = \sum_{i=1}^n \frac{(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) = 0$$

$$(ii) \sum_{i=1}^n \omega_i^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$b = \beta + \sum_{i=1} \omega_i(\varepsilon_i - \bar{\varepsilon}) = \beta + \sum_{i=1} \omega_i \varepsilon_i - \bar{\varepsilon} \sum_{i=1} \omega_i = \beta + \sum_{i=1} \omega_i \varepsilon_i \quad (11)$$

From (11) we see that b as a linear function of the error term. Taking equation (11) we can show that our OLS estimator is unbiased.

4.2.1 Unbiasedness

Consider the slope coefficient, b :

$$E(b|x) = E(\beta|x) + E\left(\sum_{i=1} \omega_i(\varepsilon_i - \bar{\varepsilon})|x\right)$$

as β is a constant, $E(\beta|x) = \beta$

$$E(b|x) = \beta + E\left(\sum_{i=1} \omega_i(\varepsilon_i - \bar{\varepsilon})|x\right) = \beta + E(\omega_1(\varepsilon_1 - \bar{\varepsilon}) + \omega_2(\varepsilon_2 - \bar{\varepsilon}) + \dots + \omega_n(\varepsilon_n - \bar{\varepsilon})|x)$$

As ω_i is made up of x_i , (then $E(\omega_i(\varepsilon_i - \bar{\varepsilon})|x_i) = \omega_i E(\varepsilon_i - \bar{\varepsilon}|x_i)$, consequently:

$$E(b|x) = \beta + \omega_1 E((\varepsilon_1 - \bar{\varepsilon})|x) + \omega_2 E((\varepsilon_2 - \bar{\varepsilon})|x) + \dots + \omega_n E((\varepsilon_n - \bar{\varepsilon})|x) = \beta$$

as $E(\varepsilon_i - \bar{\varepsilon}|x) = E(\varepsilon_i|x) - E(\bar{\varepsilon}|x) = 0 \forall i$, from CLRM assumption 1.

Therefore the slope coefficient is unbiased estimator, that is, $E(b|x) = \beta$, that is, in repeated regressions of y on x the average of the coefficient estimates of b will be equal to the true coefficient.

For the intercept, a , we have:

$$E(a|x) = E(\bar{y}|x) - E(b\bar{x}|x) = E(\alpha + \beta\bar{x} + \bar{\varepsilon}|x) - \bar{x}E(b|x) = \alpha + \beta\bar{x} + E(\bar{\varepsilon}|x) - \beta\bar{x} = \alpha$$

as $E(\bar{\varepsilon}|x) = 0$. Therefore the intercept is an unbiased estimator, that is, $E(a|x) = \alpha$. To try and give the concept of unbiasedness some meaning, consider the (unrealistic) case in which we know the true underlying relationship:

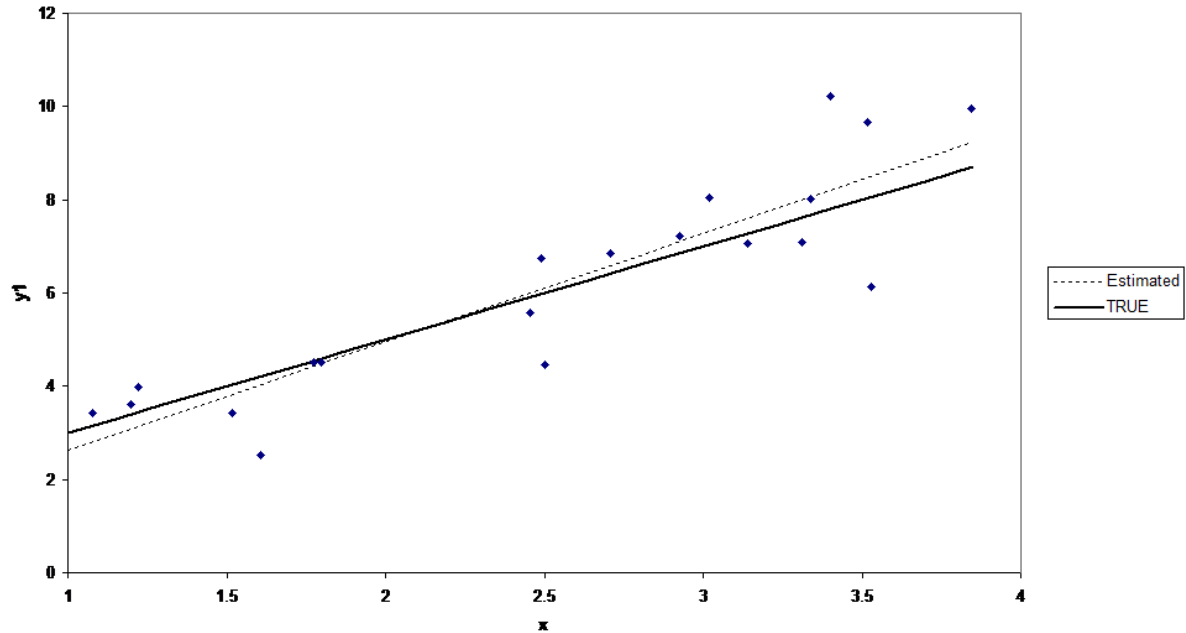
$$y_i|x_i = \alpha + \beta x_i + \varepsilon_i \quad (12)$$

where $\alpha = 1$ and $\beta = 2$.

We then observe some realisations of y_i^1 based on the known values of x_i and some random shocks ($\varepsilon_i^1 \sim N(0, 1)$) according to equation (12) – these are plotted in Figure 9a. Applying OLS to this data yields the dashed line, which compared to the true regression line (solid line), yields an intercept which is too small ($a=0.307$) and a slope

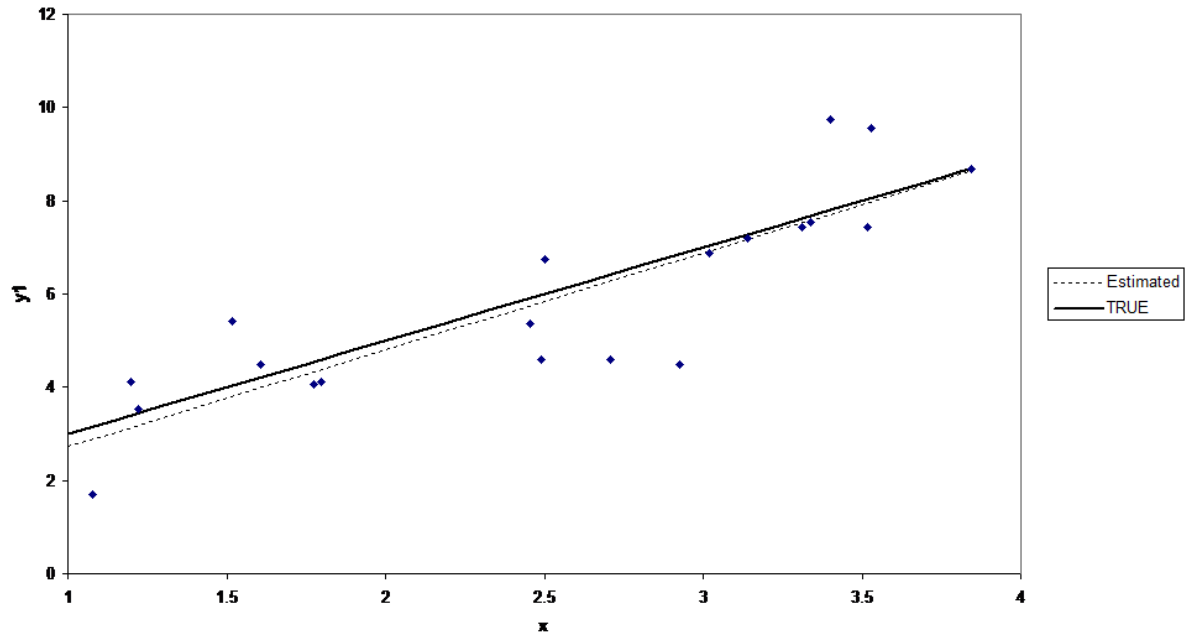
which is too large ($b=2.322$).

Figure 9a: Scatter plot of y_1 vs x

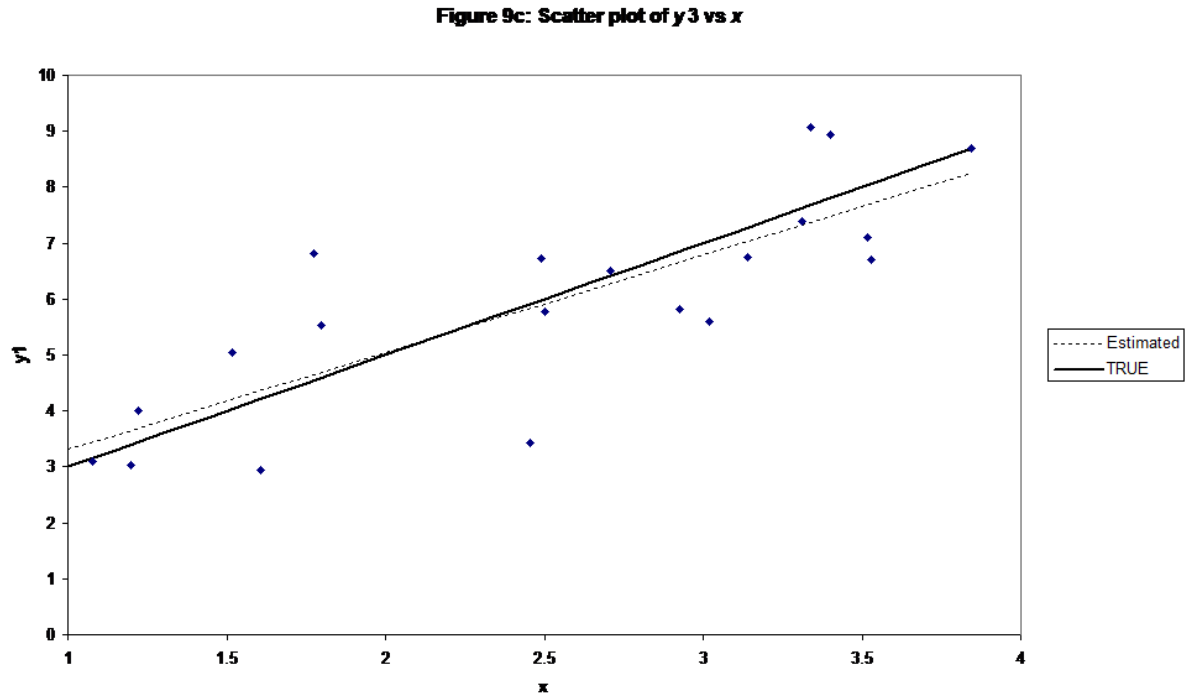


However, these realisations of y (y_i^1) are only one of an infinite number that could have arisen depending upon what shocks arose.

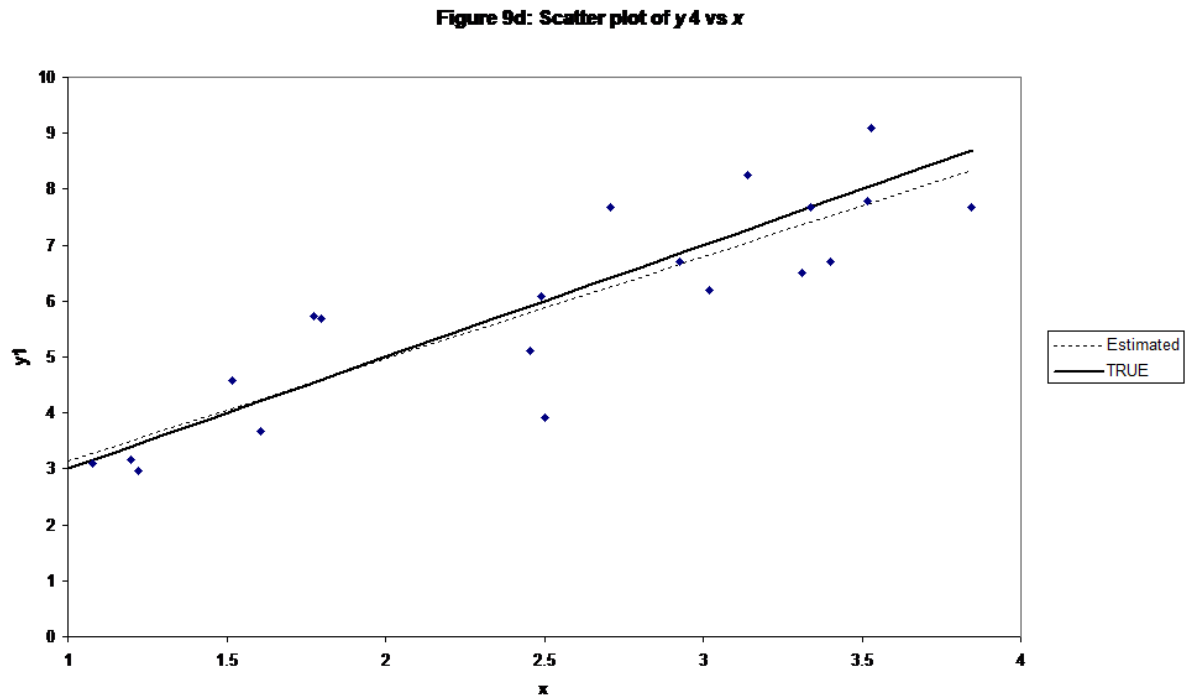
Suppose we can start the world off again and for the same values of x_i we observe some new random shocks (ε_i^2) and this yields some new values realisations of y (y_i^2) according to equation (12). These new data are plotted in Figure 9b. Applying OLS to this new data yields the dashed line, which compared to the true regression line (solid line), yields an intercept which is slightly too small ($a=0.651$) and a slope which is slightly too large ($b=2.077$).

Figure 9b: Scatter plot of y_2 vs x 

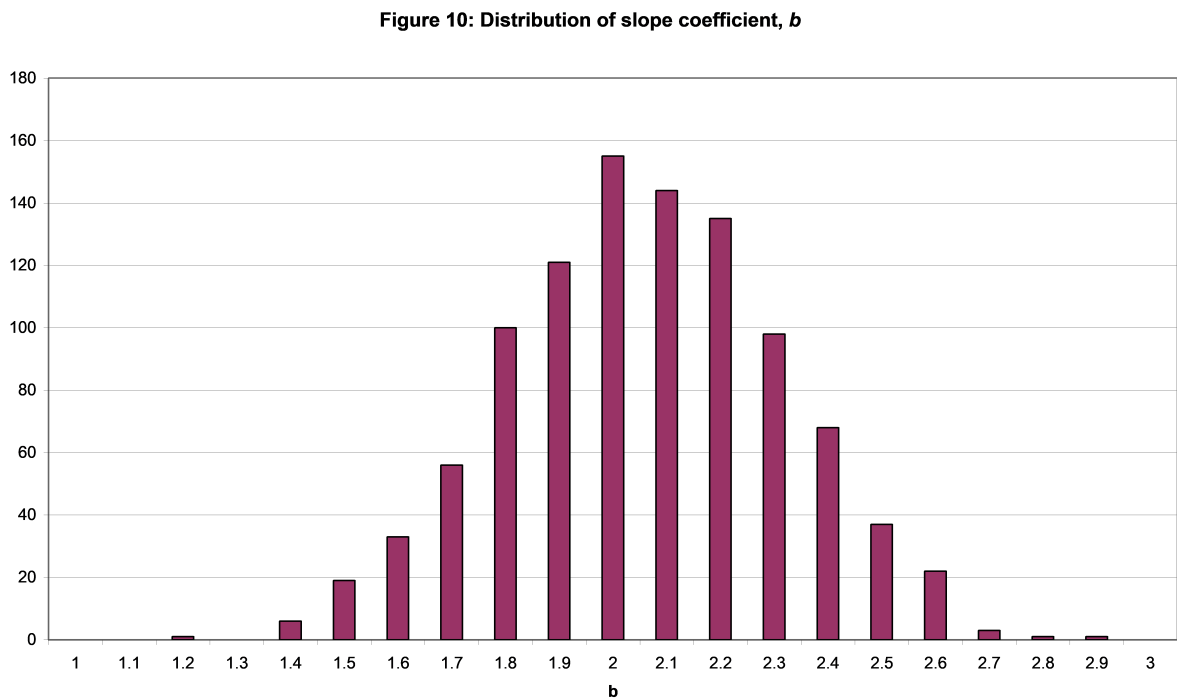
We start the world off again and for the same values of x_i (again) we observe some new random shocks (ε_i^3) and this yields some new values realisations of y (y_i^3) according to equation (12). These new data are plotted in Figure 9c. Applying OLS to this new data yields the dashed line, which compared to the true regression line (solid line), yields an intercept which is too large ($a=1.570$) and a slope which is too small ($b=1.738$).



We start the world off again and for the same values of x_i (again) we observe some new random shocks (ε_i^4) and this yields some new values realisations of y (y_i^4) according to equation (12). These new data are plotted in Figure 9d. Applying OLS to this new data yields the dashed line, which compared to the true regression line (solid line), yields an intercept which is too large ($a=1.315$) and a slope which is too small ($b=1.825$).



Imagine we start the world off 1000 times, we would get 1000 estimates of a and 1000 estimates of b . We plot the distribution of b as a histogram (Figure 10), we see that the mean of the distribution of b is 2.007 ($\simeq \beta$ and is unbiased).



4.2.2 Variances

Looking at the variance of the slope coefficient, b ,

$$V(b|x) = E[(b|x - E(b|x))^2] = E[(b|x - \beta)^2] = E \left[\left(\sum_{i=1} \omega_i \varepsilon_i | x \right)^2 \right] \quad (13)$$

$$\begin{aligned} V(b|x) &= E[(\omega_1 \varepsilon_1 + \omega_2 \varepsilon_2 + \dots + \omega_n \varepsilon_n | x)^2] \\ &= E(\omega_1^2 \varepsilon_1^2 + \omega_2^2 \varepsilon_2^2 + \dots + \omega_n^2 \varepsilon_n^2 \\ &\quad + 2\omega_1 \omega_2 \varepsilon_1 \varepsilon_2 + 2\omega_1 \omega_3 \varepsilon_1 \varepsilon_3 + \dots + 2\omega_1 \omega_n \varepsilon_1 \varepsilon_n \\ &\quad + 2\omega_2 \omega_3 \varepsilon_2 \varepsilon_3 + \dots + 2\omega_2 \omega_n \varepsilon_2 \varepsilon_n + \dots + 2\omega_{n-1} \omega_n \varepsilon_{n-1} \varepsilon_n | x) \end{aligned}$$

As ω_i^2 is made up of x_i , then $E(\omega_i^2 \varepsilon_i^2 | x_i) = \omega_i^2 E(\varepsilon_i^2 | x_i)$, consequently:

$$\begin{aligned} V(b) &= \omega_1^2 E(\varepsilon_1^2 | x) + \omega_2^2 E(\varepsilon_2^2 | x) + \dots + \omega_n^2 E(\varepsilon_n^2 | x) \\ &\quad + 2\omega_1 \omega_2 E(\varepsilon_1 \varepsilon_2 | x) + 2\omega_1 \omega_3 E(\varepsilon_1 \varepsilon_3 | x) + \dots + 2\omega_1 \omega_n E(\varepsilon_1 \varepsilon_n | x) \\ &\quad + 2\omega_2 \omega_3 E(\varepsilon_2 \varepsilon_3 | x) + \dots + 2\omega_2 \omega_n E(\varepsilon_2 \varepsilon_n | x) + \dots + 2\omega_{n-1} \omega_n E(\varepsilon_{n-1} \varepsilon_n | x) \end{aligned}$$

As $V(\varepsilon_i | x) = E(\varepsilon_i^2 | x) = \sigma^2$ and $cov(\varepsilon_i, \varepsilon_j | x) = E(\varepsilon_i \varepsilon_j | x) = 0$, $i \neq j$, all of the cross-product terms are zero and we have

$$V(b|x) = \omega_1^2 \sigma^2 + \omega_2^2 \sigma^2 + \dots + \omega_n^2 \sigma^2 = \sigma^2 \sum_{i=1} \omega_i^2 = \frac{\sigma^2}{\sum_{i=1} (x_i - \bar{x})^2} \quad (14)$$

Now we want the variance of the intercept, a ,

$$V(a|x) = V(\bar{y} - b\bar{x} | x) = V(\bar{y} | x) + \bar{x}^2 V(b|x) - 2\bar{x} cov(\bar{y}, b | x)$$

We know,

$$V(\bar{y} | x) = V(\alpha + \beta \bar{x} + \bar{\varepsilon} | x) = V(\bar{\varepsilon} | x) = \sigma^2 / n$$

and $cov(\bar{y}, b | x) = 0$ (see Appendix B). Therefore,

$$V(a|x) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1} (x_i - \bar{x})^2} \right] = \sigma^2 \left[\frac{\sum_{i=1} x_i^2 - n\bar{x}^2 + n\bar{x}^2}{n \sum_{i=1} (x_i - \bar{x})^2} \right] = \sigma^2 \left[\frac{\sum_{i=1} x_i^2}{n \sum_{i=1} (x_i - \bar{x})^2} \right] \quad (15)$$

Returning to our example in equation (12) the theoretical variance of b using equation (14) is $\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{1}{15.083} = 0.065$, compared to the estimated variance from the 1000 estimates of b (plotted in Figure 10) which is $V(b) = 0.066$. Appendix G also shows an

empirical calculation of these variances.

In addition, it is possible to work out the $cov(a, b|x) = -\frac{\sigma^2 \bar{x}}{\sum_{t=1}^n (x_t - \bar{x})^2}$ (see Appendix C for a proof of this).

Note: Information regarding the variances and covariances can be presented in a variance-covariance matrix as:

$$\begin{bmatrix} V(a|x) & cov(a, b|x) \\ cov(a, b|x) & V(b|x) \end{bmatrix}$$

where $cov(a, b|x) = cov(b, a|x)$.

As both b and a are a linear function of the error term, ε , (see equation (10)) and as ε is normally distributed (by assumption), then both a and b will follow a normal distribution (as can be seen in Figure 10 the distribution of b is Normal).

$$b|x \sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

,

$$a|x \sim N\left(\alpha, \sigma^2 \left[\frac{\sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} \right]\right)$$

Additionally as the error variance, s^2 , is the sum of squared normally distributed residuals, then

$$\frac{(n-2)s^2}{\sigma^2} \sim \chi_{n-2}^2$$

4.3 Hypothesis testing

All hypothesis testing follows a 5-step procedure:

1. $H_0 : \beta = \beta_0$
2. $H_1 : \beta \neq \beta_0$
3. Choose some appropriate significance level of c , and find the corresponding value from the t-distribution, denoted $-t_{DoF}^{c/2}$ and $t_{DoF}^{c/2}$, where DoF is the degrees of freedom of the model, that is, the number of observation, n , minus the number of restrictions on the residuals, 2, (or number of estimated parameters in the model, a and b).

4. $t = \frac{b-\beta_0}{s_b} \sim t_{DoF}$, where $s_b = \sqrt{\frac{s^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$ = standard error of b (in replacing σ^2 by s^2 in the standard error of b , we are essentially scaling a $N(0,1)$ by a $\sqrt{\chi_{DoF}^2/DoF}$ and this yields a t_{DoF} distribution.
5. If t is either less than $-t_{DoF}^{c/2}$, or greater than $t_{DoF}^{c/2}$, then we have observed an event which occurs with a probability of less than c and should therefore reject H_0 . The decision rule is: Reject H_0 if $t = \frac{b-\beta_0}{s_b} > t_{DoF}^{c/2}$; Do not reject H_0 if $t = \frac{b-\beta_0}{s_b} < t_{DoF}^{c/2}$.

4.4 Measure of goodness of fit

Does the estimated OLS model fit the data well? OLS minimises the RSS, this means that the estimates, a and b produce a smaller RSS than any alternative pair of estimates. But is the model “good” at explaining movements in y_i ? Our OLS regression has:

$$\underbrace{y_i}_{\text{Actual values}} = \underbrace{a + bx_i}_{\hat{y}_i} + \underbrace{e_i}_{\text{Residuals}}$$

taking averages

$$\bar{y} = \bar{\hat{y}} + \underbrace{\bar{e}}_0 \Rightarrow \bar{y} = \bar{\hat{y}}$$

and subtracting these two equations we have:

$$y_i - \bar{y} = (\hat{y}_i - \bar{\hat{y}}) + e_i$$

squaring both sides we get:

$$(y_i - \bar{y})^2 = (\hat{y}_i - \bar{\hat{y}})^2 + e_i^2 + 2(\hat{y}_i - \bar{\hat{y}})e_i$$

Now taking sums we have

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{TSS} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}_{ESS} + \underbrace{\sum_{i=1}^n e_i^2}_{RSS} + 2 \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})e_i}_0$$

The last term is zero, as from equation (5) we have $\sum_{i=1}^n x_i e_i = 0$ in which case: $\sum_{i=1}^n \hat{y}_i e_i$. Consequently, we have: $TSS=ESS+RSS$, where TSS = Total sum of squares (the total amount of variation in the dependent variable, y), ESS = Explained sum of squares (the amount of variation the model explained, i.e. the amount of variation in $\hat{y}$ and

RSS= Residual sum of squares (amount of variation the model did NOT explain. i.e the amount of variation in e).

Our measure of goodness of fit, is the coefficient of determination or R^2 and this is calculated as:

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{ESS}{TSS}$$

and so it can be interpreted as the proportion of the total variation in the dependent variable that the model (x_i) can explain.

Use of R^2

- 1 Goodness of fit measure for simple regression model
 - 2 It can be used to compare (choose) between different linear models (as long as the dependent variable is identical in all models).
 - 3 It is NOT correct to say that a model with a “high” R^2 is a good model.
- (see Appendix E for why in the 2-variable regression $R^2 = \text{corr}(x, y)^2$. Appendix F shows some labelled Stata output and Appendix G has an empirical example for calculating R^2).

4.5 Model adequacy

A better way to detect model adequacy than the use of R^2 is to ensure that the residuals, e_i , are consistent with the assumptions that you make about the disturbance term, ε_i . That is, we would require that:

- 1 Testing the the error term is unrelated to x_i (CLRM assumption 1).
- 2 The residuals have a constant error variance (CLRM assumption 2).
- 3 The residuals are serially uncorrelated (CLRM assumption 3).
- 4 The residuals are normally distributed (CLRM assumption 4).

We will look at these issues later in the module. (see Handout 5 onwards).

4.6 Forecasting/Prediction

One important use of regression is the computation of predictions, or forecasts, for the dependent variable, conditional on an assumed value for the explanatory variable. Suppose that the explanatory variable is equal to x_{n+1} and that the linear relationship continues to hold, then:

$$y_{n+1} = \alpha + \beta x_{n+1} + \varepsilon_{n+1}$$

Often we are interested in forecasting the actual value that will result for y_{n+1} . Our estimator is written:

$$\hat{y}_{n+1} = a + bx_{n+1}$$

and we can think of defining the prediction/forecast error as $e_{n+1} = y_{n+1} - \hat{y}_{n+1}$. Where

$$E(e_{n+1}|x) = E(y_{n+1} - \hat{y}_{n+1}|x) = E[(\alpha - a) + (\beta - b)x_{n+1} + \varepsilon_{n+1}|x]$$

$$E(e_{n+1}|x) = E(y_{n+1} - \hat{y}_{n+1}|x) = E(\alpha - a|x) + x_{n+1}E(\beta - b|x) + E(\varepsilon_{n+1}|x) = 0$$

The variance of the prediction error $V(e_{n+1}|x) = V(y_{n+1} - \hat{y}_{n+1}|x)$ is:

$$V(e_{n+1}|x) = V(y_{n+1} - \hat{y}_{n+1}|x) = \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_{t=1}^n (x_t - \bar{x})^2} \right]$$

(see Appendix D for more details on calculating this variance term).

Given that $\varepsilon_{n+1}|x \sim N(0, \sigma^2)$, we can write that

$$e_{n+1}|x \sim N \left(0, \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_{t=1}^n (x_t - \bar{x})^2} \right] \right)$$

and as σ^2 is unknown and as we use of s^2 in its place, this means the distribution will be: $\frac{e_{n+1}-0}{\hat{V}(e_{n+1})} \sim t_{n-2}$ and hence

$$P(-t_{n-2}^{c/2} \leq \frac{y_{n+1} - \hat{y}_{n+1}}{\sqrt{\hat{V}(e_{n+1})}} \leq t_{n-2}^{c/2}) = 1-c \Rightarrow P(-t_{n-2}^{c/2} \sqrt{\hat{V}(e_{n+1})} \leq y_{n+1} - \hat{y}_{n+1} \leq t_{n-2}^{c/2} \sqrt{\hat{V}(e_{n+1})})$$

Or arranging further:

$$P(\hat{y}_{n+1} - t_{n-2}^{c/2} \sqrt{\hat{V}(e_{n+1})} \leq y_{n+1} \leq \hat{y}_{n+1} + t_{n-2}^{c/2} \sqrt{\hat{V}(e_{n+1})}) = 1 - c$$

i.e. the $(1 - c)\%$ CI for y_{n+1} can be written:

$$\hat{y}_{n+1} \pm t_{n-2}^{c/2} \sqrt{s^2 \left[1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]}$$

and the CI is wider the further x_{n+1} is from $\bar{x}$. Appendix G works through an example in which we calculate the 95% confidence interval for our forecast of y_{n+1} .

5 Interpreting coefficients

OLS is appropriate for all model that are linear in parameters and each of the following models is linear in parameters (although the model maybe non-linear in the variables y and x):

1. Given the model $y_i = \alpha + \beta x_i + \varepsilon_i$, we have:

$$\frac{\partial y_i}{\partial x_i} = \beta = \frac{\text{Change in } y_i}{1 \uparrow \text{ in } x_i}$$

2. Given the model $y_i = \alpha + \beta \ln(x_i) + \varepsilon_i$, we have:

$$\frac{\partial y_i}{\partial \ln(x_i)} = \beta = \frac{\text{Change in } y_i}{1 \uparrow \text{ in } \ln(x_i)}$$

Digression

$$\frac{\partial \ln(x)}{\partial x} = \frac{1}{x} \Rightarrow \partial \ln(x) = \frac{\partial x}{x} = \text{Proportionate change in } x$$

Alternatively, define the growth rate of the variable x as g ,

$$g = \frac{x^+ - x}{x} = \frac{x^+}{x} - 1 \Rightarrow 1 + g = \frac{x^+}{x}$$

Now taking nature logs of both sides we get: $\ln(1+g) = \ln(x_t/x_{t-1}) = \ln(x^+) - \ln(x) \equiv \Delta \ln(x) \approx g$ as:

g	$\ln(1+g)$	g	$\ln(1+g)$	g	$\ln(1+g)$
0.01	0.0099	0.10	0.0953	-0.03	-0.0305
0.02	0.0198	0.20	0.1823	-0.06	-0.0619
0.03	0.0296	-0.01	-0.0101	-0.10	-0.1054
0.06	0.0583	-0.02	-0.0202	-0.20	-0.2231

But what is a unit increase in $\ln(x)$?

$$\ln(x^+/x) = 1 \Rightarrow x^+/x = \exp(1) = 2.718 \Rightarrow g = 1.718$$

$$\frac{\partial y_i}{\partial \ln(x_i)} = \beta = \frac{\text{Change in } y_i}{171.8\% \uparrow \text{ in } x_i}$$

However, if we increase $\ln(x)$ by 0.01 then

$$\frac{\beta}{100} = \frac{\text{Change in } y_i}{1\% \uparrow \text{ in } x_i}$$

Alternatively consider our equation for some value x^0

$$E(y^0|x^0) = \alpha + \beta \ln(x^0)$$

Which then increases to x^1

$$E(y^1|x^1) = \alpha + \beta \ln(x^1)$$

Subtracting these equations we have:

$$E(y^1|x^1) - E(y^0|x^0) = \beta \ln(x^1) - \beta \ln(x^0) = \beta \ln(x^1/x^0)$$

And hence $\beta \ln(x^1/x^0) = \beta g$ is the expected change in y when

$$g = \frac{x_1 - x_0}{x_0} = \frac{x_1}{x_0} - 1 \Rightarrow 1 + g = \frac{x_1}{x_0} \Rightarrow \ln(x^1/x^0) = \ln(1 + g) = g$$

(if g is small).

3. Given the model $\ln(y_i) = \alpha + \beta x_i + \varepsilon_i$, we have

$$\frac{\partial \ln(y_i)}{\partial x_i} = \beta \approx \frac{\text{Proportionate change in } y_i}{1 \uparrow \text{ in } x_i}$$

and

$$100\beta \approx \frac{\% \text{ change in } y_i}{1 \uparrow \text{ in } x_i}$$

or

$$100(\exp(\beta) - 1) = \frac{\% \text{ change in } y_i}{1 \uparrow \text{ in } x_i}$$

4. Given the model $\ln(y_i) = \alpha + \beta \ln(x_i) + \varepsilon_i$, we have

$$\frac{\partial \ln(y_i)}{\partial \ln(x_i)} = \beta = \frac{\text{Proportionate change in } y_i}{1 \text{ proportionate } \uparrow \text{ in } x_i} = \frac{\% \text{ change in } y_i}{1\% \uparrow \text{ in } x_i}$$

Appendices

A Rules on expectations/variances

Define $E(w) = \sum_{i=1}^k p_i w_i$ as the expected value of the random variable w and $V(w) = E[w - E(w)]^2 = E(w^2) - E(w)^2 = \sum_{i=1}^k p_i (w_i - E(w))^2$. Then:

1. $E(a + x) = \sum_{i=1}^k p_i (a + x_i) = a + \sum_{i=1}^k p_i x_i = a + E(x)$.
2. $E(ax) = \sum_{i=1}^k p_i (ax_i) = a \sum_{i=1}^k p_i x_i = aE(x)$.
3. $V(a + X) = \sum_{i=1}^k p_i [(a + x_i) - E(a + X)]^2 = \sum_{i=1}^k p_i [(a + x_i) - a - E(X)]^2$
 $= \sum_{i=1}^k p_i [x_i - E(X)]^2 = V(X)$
4. $V(aX) = \sum_{i=1}^k p_i [ax_i - E(aX)]^2 = \sum_{i=1}^k p_i [ax_i - aE(X)]^2$
 $= a^2 \sum_{i=1}^k p_i [x_i - E(X)]^2 = a^2 V(X)$
5. $cov(a + x, y) = cov(x, y)$
6. $cov(ax, y) = acov(x, y)$.
7. $V(x + y) = V(x) + V(y) + 2cov(x, y)$.
8. $V(x - y) = V(x) + V(y) - 2cov(x, y)$.
9. $V(x + y + z) = V(x) + V(y) + V(z) + 2cov(x, y) + 2cov(x, z) + 2cov(y, z)$.
10. $V(x - y - z) = V(x) + V(y) + V(z) - 2cov(x, y) - 2cov(x, z) + 2cov(y, z)$.

B Showing $cov(\bar{y}, \bar{x}b|x) = 0$

In deriving $V(a|x)$ it is stated $cov(\bar{y}, \bar{x}b|x) = 0$. In this appendix we show this is the case, where we will be using the following observations:

- $b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2}$
- $\sum(x_i - \bar{x}) = 0$
- $E(y_i) = \mu_y$
- $V(y) = E(y_i^2) - E(y_i)^2 = \sigma_y^2 \rightarrow E(y_i^2) = \sigma_y^2 + \mu_y^2$
- $E(y_i \cdot y_j) = E(y_i)E(y_j) = \mu_y^2 \quad i \neq j$

We drop the conditional on x in the following for ease of notation.

$$\begin{aligned} cov(\bar{y}, \bar{x}b) &= cov\left[\frac{\sum y_i}{n}, \bar{x} \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2}\right] \\ &= \frac{\bar{x}}{n \sum(x_i - \bar{x})^2} cov\left[\sum y_i, \sum(x_i - \bar{x})y_i\right] \end{aligned}$$

Defining $d = \frac{\bar{x}}{n \sum(x_i - \bar{x})^2}$ we can write:

$$cov(\bar{y}, \bar{x}b) = d cov\left[\sum y_i, \sum(x_i - \bar{x})y_i\right]$$

Given we define $cov(x, y) = E(x \cdot y) - E(x) \cdot E(y)$

$$\begin{aligned} cov(\bar{y}, \bar{x}b) &= d \left(E\left[\sum y_i \cdot \sum(x_i - \bar{x})y_i\right] - E\left[\sum y_i\right] \cdot E\left[\sum(x_i - \bar{x})y_i\right] \right) \\ &= d \left(E\left[\sum y_i \cdot \sum(x_i - \bar{x})y_i\right] - \sum E(y_i) \cdot \sum(x_i - \bar{x})E(y_i) \right) \\ &= d \left(E\left[\sum y_i \cdot \sum(x_i - \bar{x})y_i\right] - \sum \mu_y \cdot \sum(x_i - \bar{x})\mu_y \right) \\ &= d \left(E\left[\sum y_i \cdot \sum(x_i - \bar{x})y_i\right] - n\mu_y \cdot \mu_y \sum(x_i - \bar{x}) \right) \\ &= d E\left[\sum y_i \cdot \sum(x_i - \bar{x})y_i\right] \end{aligned}$$

$$\begin{aligned}
\text{cov}(\bar{y}, \bar{x}b) &= d \, E \left[\sum_{i=1}^n (x_i - \bar{x}) y_i^2 + \sum_{i \neq j} (x_i - \bar{x}) y_i y_j \right] \\
&= d \left[\sum_{i=1}^n (x_i - \bar{x}) E(y_i^2) + \sum_{i \neq j} (x_i - \bar{x}) E(y_i y_j) \right] \\
&= d \left[\sum_{i=1}^n (x_i - \bar{x}) (\sigma_y^2 + \mu_y^2) + \sum_{i \neq j} (x_i - \bar{x}) E(y_i) E(y_j) \right] \\
&= d \left[(\sigma_y^2 + \mu_y^2) \left[\sum_{i=1}^n (x_i - \bar{x}) \right] + \sum_{i \neq j} (x_i - \bar{x}) \mu_y \mu_y \right] \\
&= d \left[(\sigma_y^2 + \mu_y^2) \left[\sum_{i=1}^n (x_i - \bar{x}) \right] + \mu_y^2 \sum_{i \neq j} (x_i - \bar{x}) \right] \\
&= d (\sigma_y^2 + 2\mu_y^2) \left[\sum_{i=1}^n (x_i - \bar{x}) \right] = 0
\end{aligned}$$

C Calculating $cov(a, b)$

$$cov(a, b) = E[(a - E(a))(b - E(b))] = E[(a - \alpha)(b - \beta)]$$

Now from equation (6) we have $a = \bar{y} - b\bar{x}$ and from equation (11) we have $b = \beta + \sum_{i=1} \omega_i \varepsilon_i$. Therefore:

$$b - \beta = \sum_{i=1} \omega_i \varepsilon_i$$

and given

$$\bar{y} = \alpha + \beta\bar{x} + \bar{\varepsilon}$$

we can rewrite equation (6) our equation for a as

$$a = \alpha + \beta\bar{x} + \bar{\varepsilon} - b\bar{x} = \alpha - (b - \beta)\bar{x}$$

This implies:

$$a - \alpha = -(b - \beta)\bar{x}$$

In which case, we can write:

$$E[(a - \alpha)(b - \beta)] = E[-(b - \beta)\bar{x}(b - \beta)] = -\bar{x}E[(b - \beta)^2]$$

$$E[(a - \alpha)(b - \beta)] = -\bar{x}E\left[\left(\sum_{i=1} \omega_i \varepsilon_i\right)^2\right]$$

From equation (13) and (14) we know:

$$E\left[\left(\sum_{i=1} \omega_i \varepsilon_i\right)^2\right] = \frac{\sigma^2}{\sum_{i=1} (x_i - \bar{x})^2}$$

in which case:

$$cov(a, b) = E[(a - \alpha)(b - \beta)] = \frac{-\bar{x}\sigma^2}{\sum_{i=1} (x_i - \bar{x})^2}$$

.

D Calculating $V(y_{n+1} - \hat{y}_{n+1})$

$$\begin{aligned} V(y_{n+1} - \hat{y}_{n+1}) &= V(y_{n+1}) + V(\hat{y}_{n+1}) - 2\text{cov}(y_{n+1}, \hat{y}_{n+1}) \\ &= V(\alpha + \beta x_{n+1} + \varepsilon_{n+1}) + V(\hat{y}_{n+1}) - 2\text{cov}(\alpha + \beta x_{n+1} + \varepsilon_{n+1}, \hat{y}_{n+1}) \end{aligned}$$

the last term is zero as $(\alpha + \beta x_{n+1})$ is a constant and the ε_{n+1} is unrelated to everything. Additionally, we know that

$$V(\alpha + \beta x_{n+1} + \varepsilon_{n+1}) = V(\varepsilon_{n+1}) = \sigma^2$$

In which case:

$$\begin{aligned} V(y_{n+1} - \hat{y}_{n+1}) &= \sigma^2 + V(\overbrace{\alpha + \beta x_{n+1}}^{\hat{y}_{n+1}}) = \sigma^2 + [V(a) + x_{n+1}^2 V(b) + 2x_{n+1} \text{cov}(a, b)] \\ &= \sigma^2 + \frac{\sigma^2 \sum_i x_i^2 / n}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sigma^2 x_{n+1}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} - 2 \frac{\sigma^2 x_{n+1} \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

Taking out the common factor of $\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$ from the three terms we get:

$$V(y_{n+1} - \hat{y}_{n+1}) = \sigma^2 + \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \left[x_{n+1}^2 - 2x_{n+1} \bar{x} + \sum_i x_i^2 / n \right]$$

Adding in (and subtracting out) the term $\frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$, we get:

$$V(y_{n+1} - \hat{y}_{n+1}) = \sigma^2 + \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \left[x_{n+1}^2 - 2x_{n+1} \bar{x} + \bar{x}^2 - \bar{x}^2 + \sum_{i=1}^n x_i^2 / n \right]$$

The first three terms in $\square$ can be written more succinctly as $(x_{n+1} - \bar{x})^2$ and the last 2 terms are $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ so we can write the $V(y_{n+1} - \hat{y}_{n+1})$ as:

$$\begin{aligned} V(y_{n+1} - \hat{y}_{n+1}) &= \sigma^2 + \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \left[(x_{n+1} - \bar{x})^2 + \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \right] \\ &= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right] \end{aligned}$$

$$\mathbf{E} \quad R^2 = \text{corr}(x, y)^2$$

In the 2-variable regression model $R^2 = \text{corr}(x, y)^2$. To show this remember:

$$\text{corr}(x, y) = \frac{\text{cov}(x, y)}{\sqrt{V(x) \cdot V(y)}} \rightarrow \text{corr}(x, y)^2 = \frac{\text{cov}(x, y)^2}{V(x) \cdot V(y)}$$

We know:

$$R^2 = \frac{ESS}{TSS} = \frac{ESS}{\sum (y_i - \bar{y})^2} = \frac{ESS}{V(y)(n-1)}$$

In the two variable regression model

$$ESS = \sum (\hat{y}_i - \bar{y})^2 = b^2 \sum (x_i - \bar{x})^2$$

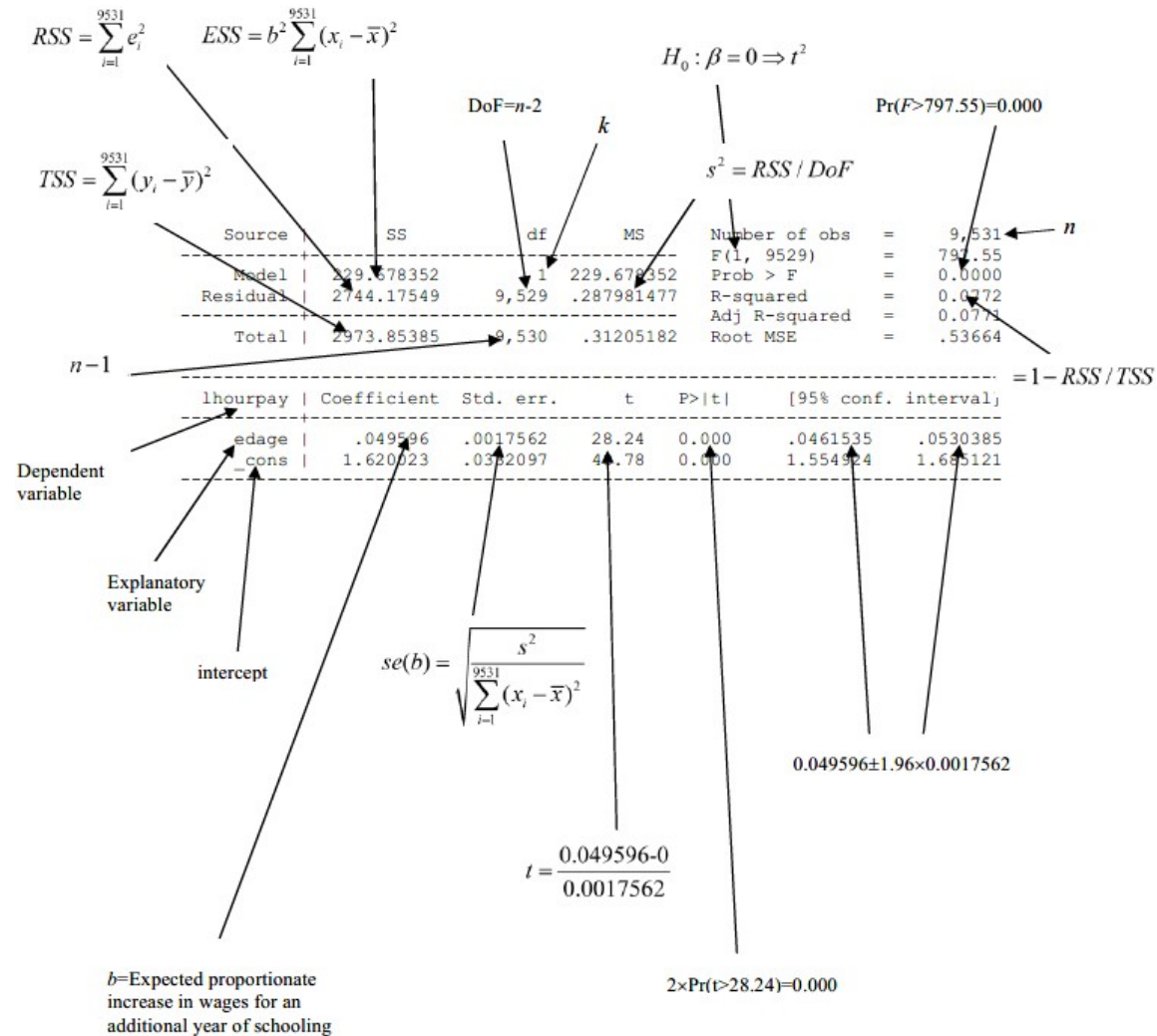
$$ESS = \left(\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \right)^2 \sum (x_i - \bar{x})^2 = \left(\frac{\text{cov}(x, y)}{V(x)} \right)^2 V(x)(n-1) = \frac{\text{cov}(x, y)^2}{V(x)}(n-1)$$

In which case:

$$R^2 = \frac{ESS}{TSS} = \frac{ESS}{V(y)(n-1)} = \frac{\frac{\text{cov}(x, y)^2}{V(x)}(n-1)}{V(y)(n-1)} = \frac{\text{cov}(x, y)^2}{V(x)V(y)} = \text{corr}(x, y)^2$$

F Stata regression output

regress lhourpay edage if(edage>= 0 & edage<= 95)



G Exercise questions

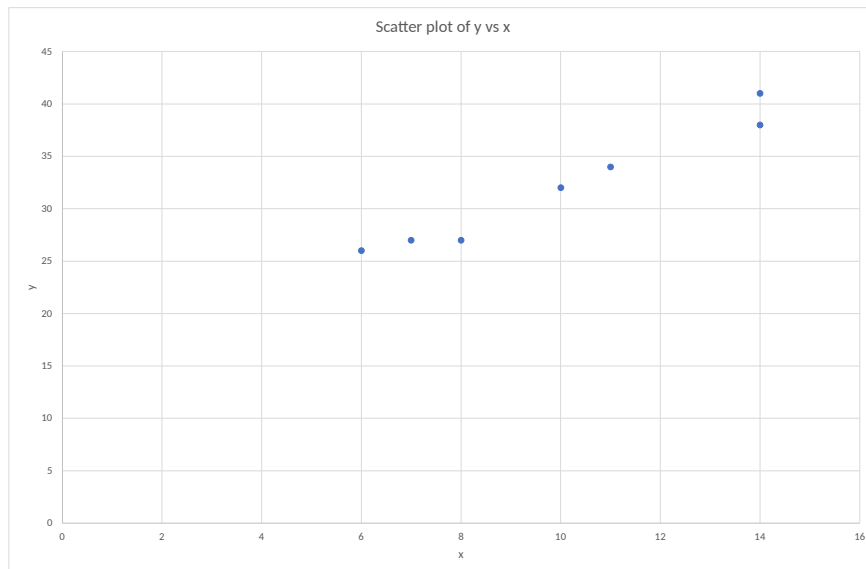
Given the data on the variables x and y :

x	7	14	8	10	11	6	14
y	27	41	27	32	34	26	38

- (a) Plot y against x .
- (b) Obtain the least squares regression line for a regression of y on x .
- (c) Calculate the residuals from the estimated regression and check that $\sum_{i=1} e_i = 0$ and $\sum_{i=1} x_i e_i = 0$.
- (d) Calculate R^2 , standard error of the regression and the standard error of the coefficient estimates.
- (e) Test the hypothesis that the slope coefficient is zero.
- (f) Given $x_8 = 15$, obtain a prediction for y_8 . Obtain a 95% confidence interval for y_8 .

Answer

(a) Scatter plot



(b)

$$\sum_i x_i = 70, \sum_i y_i = 225, \sum_i x_i^2 = 762, \sum_i y_i^2 = 7439, \sum_i x_i y_i = 2361$$

$$b = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2} = \frac{2361 - 7(10)(32.14)}{762 - 7(100)} = 1.79$$

$$a = 32.14 - 1.79(10) = 14.24$$

(c)

$$e_t = y_t - (a + bx_t) = y_t - (14.24 + 1.790x_t)$$

	1	2	3	4	5	6	7	Σ
e	0.2281	1.6959	-1.5622	-0.1429	0.0668	1.0184	-1.3042	-0.0001
$x \times e$	1.5968	23.742	-12.498	-1.4286	0.7350	6.1106	-18.258	-0.0002

(Note sums not exactly zero due to rounding in coefficient estimates for a and b).

$$(d) \text{ } RSS = 8.131; R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{8.131}{206.857} = 0.961; s = \hat{\sigma} = \sqrt{\frac{8.131}{5}} = 1.275$$

$$se(b) = \sqrt{\frac{\hat{\sigma}^2}{\sum_i x_i^2 - n\bar{x}^2}} = \frac{1.275}{\sqrt{62}} = 0.162$$

$$se(a) = \sqrt{\frac{\hat{\sigma}^2 \sum_i x_i^2}{n \left(\sum_i x_i^2 - n\bar{x}^2 \right)}} = \sqrt{\frac{1.275^2(762)}{7(62)}} = 1.689$$

(e) $H_0 : \beta = 0, H_0 : \beta \neq 0 \quad t_5^{0.025} = \pm 2.571$

$$t = \frac{1.790 - 0}{0.162} = 11.05 \Rightarrow \text{Reject } H_0$$

(f) $\hat{y}_8 = 14.24 + 1.79(15) = 41.09$

$$V(y) = s^2 \left[1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right] = 1.626 \left[1 + \frac{1}{7} + \frac{(15 - 10)^2}{62} \right] = 2.514$$

$$y_8 \in (41.09 \pm 2.571\sqrt{2.514}) \Rightarrow y_8 \in (37.01, 45.17)$$

H Stata regression commands

```
/* Read the Stata dataset qlfs18.dta from the directory C:\textbackslash Documents,
into Stata */
use "C:\Documents\qlfs18.dta"
/* Converting some variable names written in capitals*/
ren AGE age
ren EDAGE edage
ren HOURPAY hourpay
/*Plot a histogram of hourpay (with 30 bins) */
histogram hourpay, bin(30) percent
/* Due to asymmetric distribution use ln(hourpay) name the variable and plot
distribution */
gen lhourpay=ln(hourpay)
label variable lhourpay "ln(hourpay)"
histogram lhourpay, bin(30) percent
/* You are looking at the returns to education and use the variable edage
(Age when completed full time education). Tabulate this variable, it includes
individuals who have edage: "Does not apply" (-9); "No answer" (-8); "Still in
education" (96); and "Never had an education" (97). Restrict the sample to
individuals with edage values which are non-negative and below 96 */
regress lhourpay edage if(edage>=0 & edage<=95)
/* Output from this regression can be seen in Appendix F. To generate the
residuals and fitted values (predicted values) from this regression */
predict res1, resid
predict yhat1, xb
/* Worried about the influence of the extreme values in lhourpay on the results,
and trim the values of this variables by dropping the top and bottom 1% of values */
summarize lhourpay, detail
return list
scalar lhpay_1=r(p1)
scalar lhpay_99=r(p99)
regress lhourpay edage if(edage>=0 & edage<=95 & lhourpay>=lhpay_1 & /*
*/ lhourpay<=lhpay_99)
/* You can access much of the output from the regression, which would enable
you construct some statistics: number of observations (n), the r-squared (R^2),
residual sum of squares (RSS), residual degrees of freedom (DoF) and the number
of slope coefficients (k) */
```

```
scalar n=e(N)
scalar rsquared=e(r2)
scalar rss=e(rss)
scalar dof=e(df_r)
scalar k=e(df_m)
/* This saves the vector of coefficient estimates and the variance-covariance
matrix of the coefficient estimates */
matrix b=e(b)
matrix vb=e(V)
/* This is construct a t-test on the slope coefficient and the F-test that all
slope coefficients are 0 (not for a single restriction this is just the square
of the t-test */
scalar t_b=b[1,1]/sqrt(vb[1,1])
scalar f=rsquared/k/((1-rsquared)/dof)
```

I R regression commands

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)
```

```
# Read the Stata dataset qlfs18.dta from the directory C:\textbackslash Documents,
# into R
qlfs18 <- read_dta("lfsp_js18_eul_pwt18.dta")
# Converting some variable names written in capitals
names(qlfs18)[names(qlfs18) == "EDAGE"] <- "edage"
names(qlfs18)[names(qlfs18) == "HOURLPAY"] <- "hourpay"
names(qlfs18)[names(qlfs18) == "AGE"] <- "age"
#Plot a histogram of hourpay (with 30 bins)
ggplot(qlfs18, aes(x = hourpay)) +
  geom_histogram(bin=30)
# Due to asymmetric distribution use ln(hourpay) name the variable and plot
# distribution
qlfs18 <- qlfs18 %>%
  mutate(lhourpay=log(hourpay))
```

```

label1 <- c("lhourpay"="ln(hourpay)")
label(qlfs18) = as.list(label1[match(names(qlfs18),names(label1))])
ggplot(qlfs18, aes(x = lhourpay)) +
  geom_histogram(bin=30)
# You are looking at the returns to education and use the variable edage (Age
# when completed full time education). Tabulate this variable, it includes
# individuals who have edage: "Does not apply" (-9); "No answer" (-8); "Still
# in education" (96); and "Never had an education" (97). Restrict the sample
# to individuals with edage values which are non-negative and below 96
qlfs18 <- qlfs18 %>%
  mutate(edage1 = as.numeric(as.character(edage)))
model1 <- lm(lhourpay ~ edage1, data = subset(qlfs18, edage1 >=0 & edage1<=95))
summary(model1)
# Output from this regression can be seen in Appendix F. To generate the residuals
# and fitted values (predicted values) from this regression
res1 <- resid(model1)
yhat1 <- fitted(model1)
# Worried about the influence of the extreme values in lhourpay on the results,
# and trim the values of this variables by dropping the top and bottom 1% of values
x <- quantile(qlfs18$lhourpay, probs = c(0.01, 0.99), na.rm = T)
model2 <- lm(lhourpay ~ edage1, data=subset(qlfs18, edage1>=0 & edage1<=95 &
lhourpay>=x[1] & lhourpay<=x[2]))
summary(model2)
# You can access much of the output from the regression, which would enable you
# construct some statistics: number of observations (n), the r-squared (R^2),
# residual, sum of squares (RSS), residual degrees of freedom (DoF) and the
# number of slope coefficients (k)
n <- nobs(model2)
rsquared <- summary(model2)$r.squared
qlfs18 <- qlfs18 %>%
  mutate(rss = sum(res1^2))
k <- length(model2$coefficients)-1
df <- n - k - 1
# This saves the vector of coefficient estimates and the variance-covariance
# matrix of the coefficient estimates
b <- coef(model2)
vb <- vcov(model2)

```

```
# This is construct a t-test on the slope coefficient and the F-test that all  
# slope coefficients are 0 (not for a single restriction this is just the square  
# of the t-test  
t_b <- b[2]/sqrt(vb[2,2])  
f <- rsquared/k/((1-rsquared)/df)
```

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